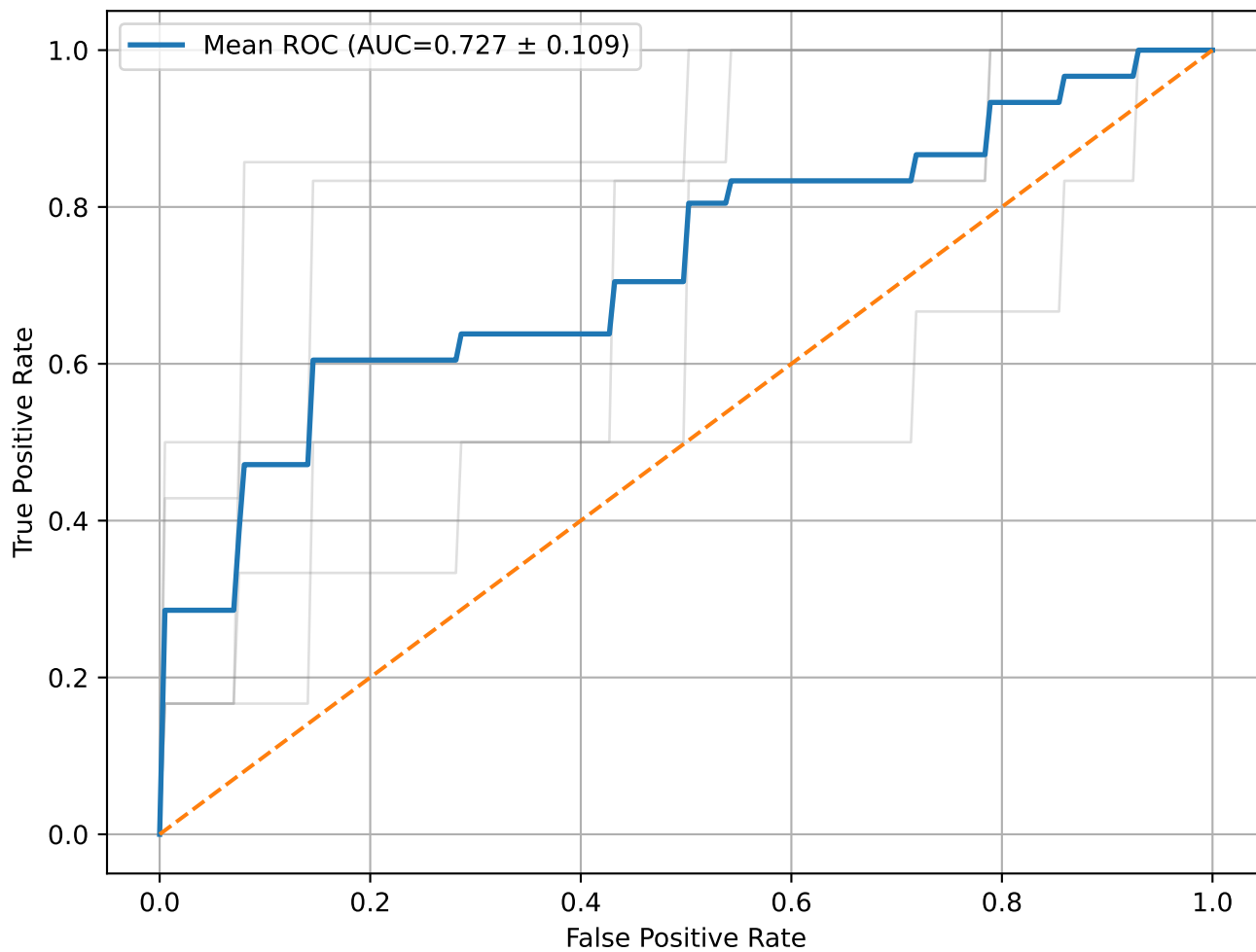


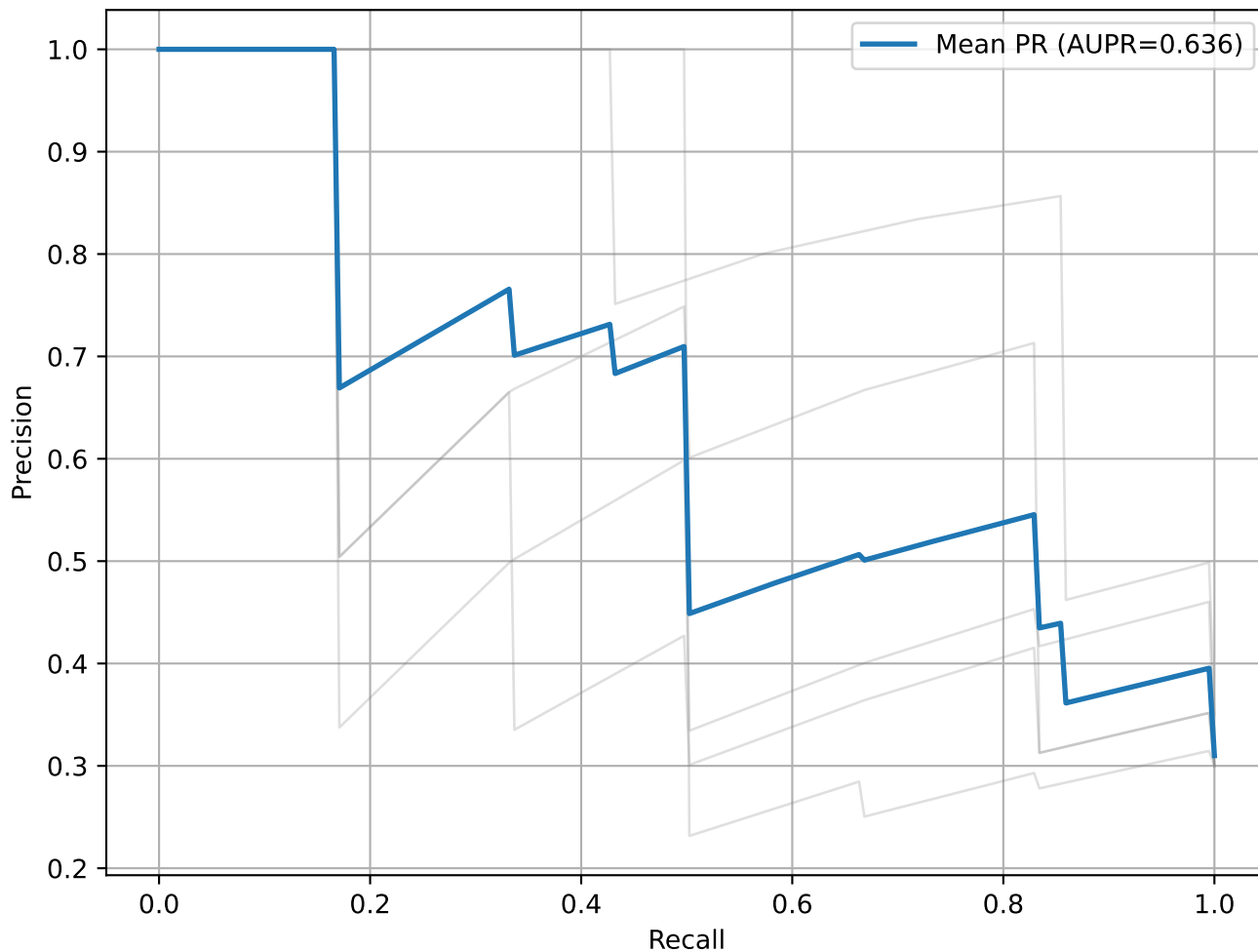
ML Models Comparison Report

Generated: 2025-12-05T16:08:32.709286
Dataset samples: 100
Models: logistic, rf, svm, xgboost, lightgbm, dnn
StratifiedKFold: 5 splits
Random seed: 42

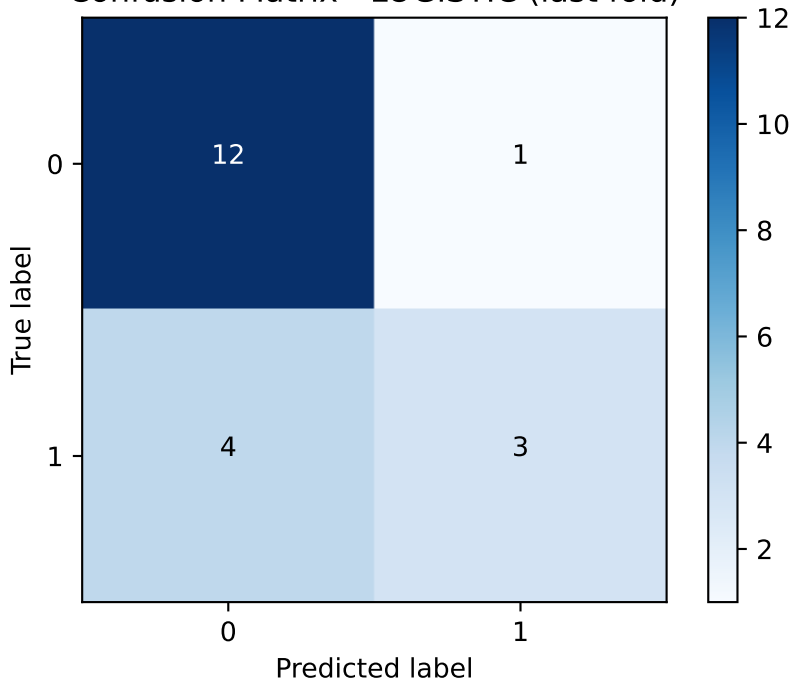
ROC - LOGISTIC



Precision-Recall - LOGISTIC



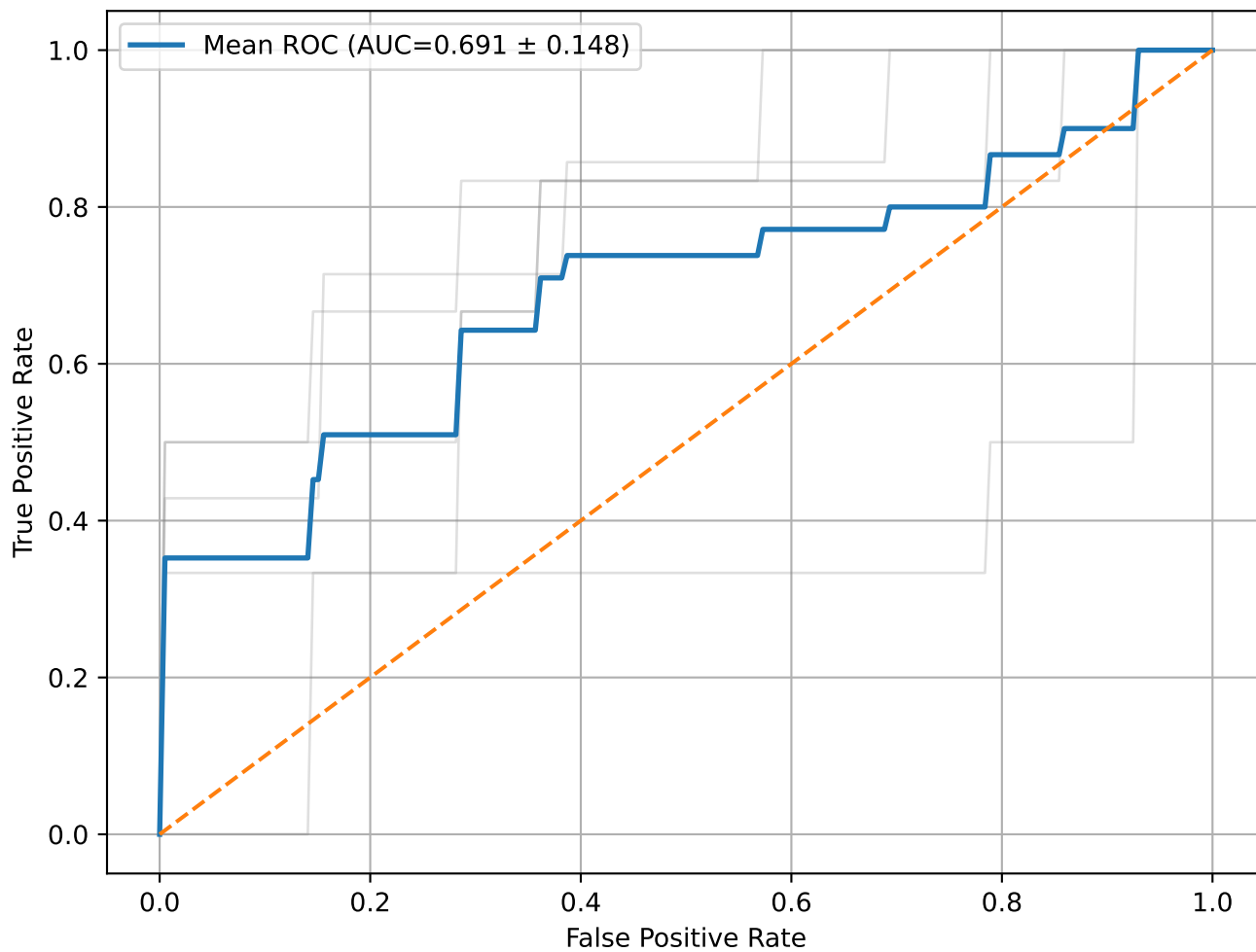
Confusion Matrix - LOGISTIC (last fold)



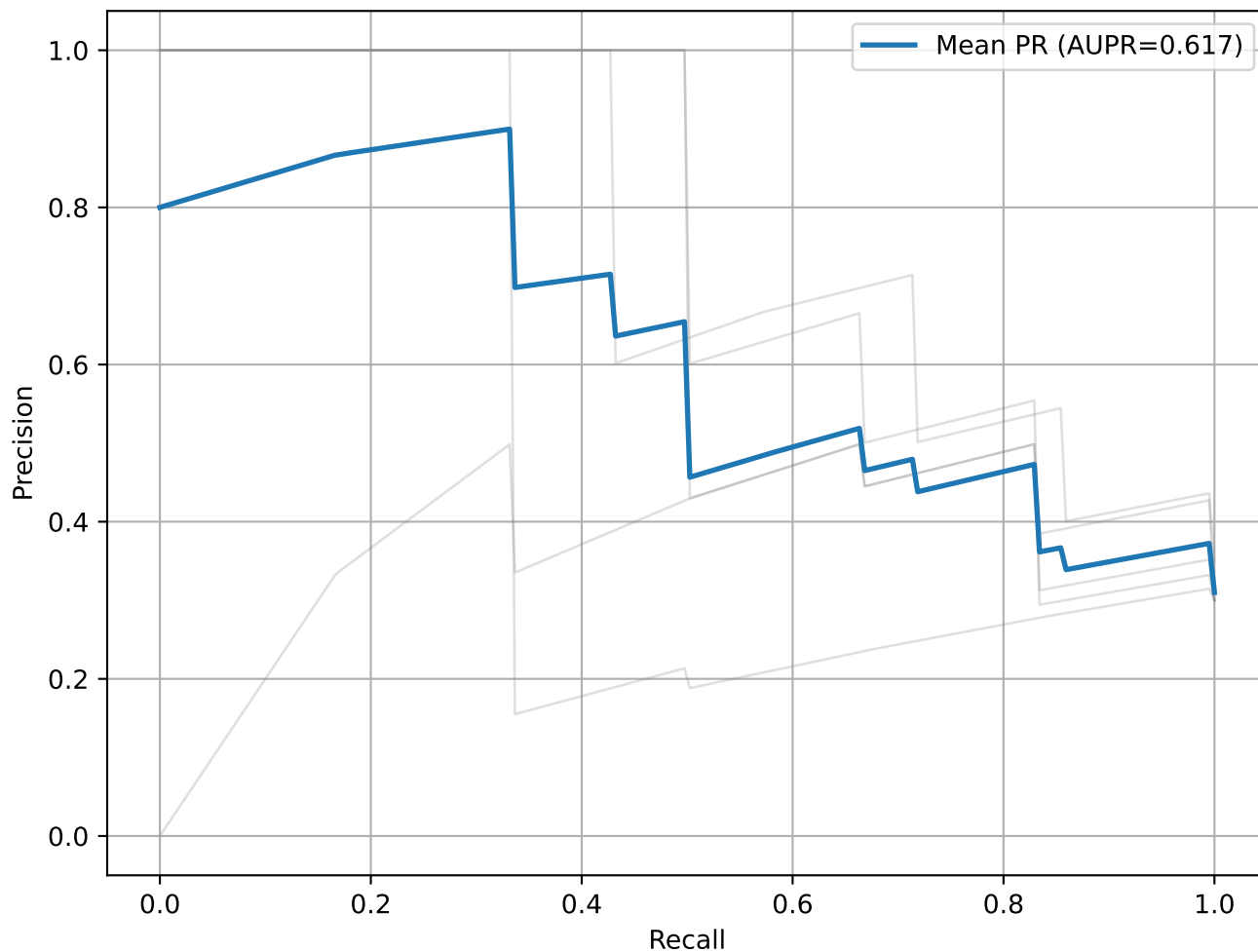
	precision	recall	f1-score	support
0	0.75	0.92	0.83	13
1	0.75	0.43	0.55	7
accuracy			0.75	20
macro avg	0.75	0.68	0.69	20
weighted avg	0.75	0.75	0.73	20

Classification Report - LOGISTIC (last fold)

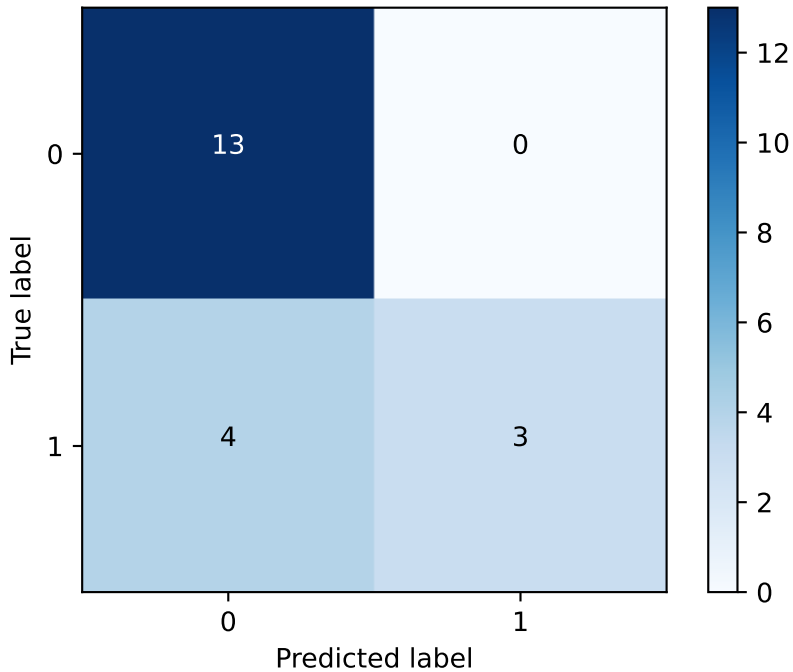
ROC - RF



Precision-Recall - RF



Confusion Matrix - RF (last fold)



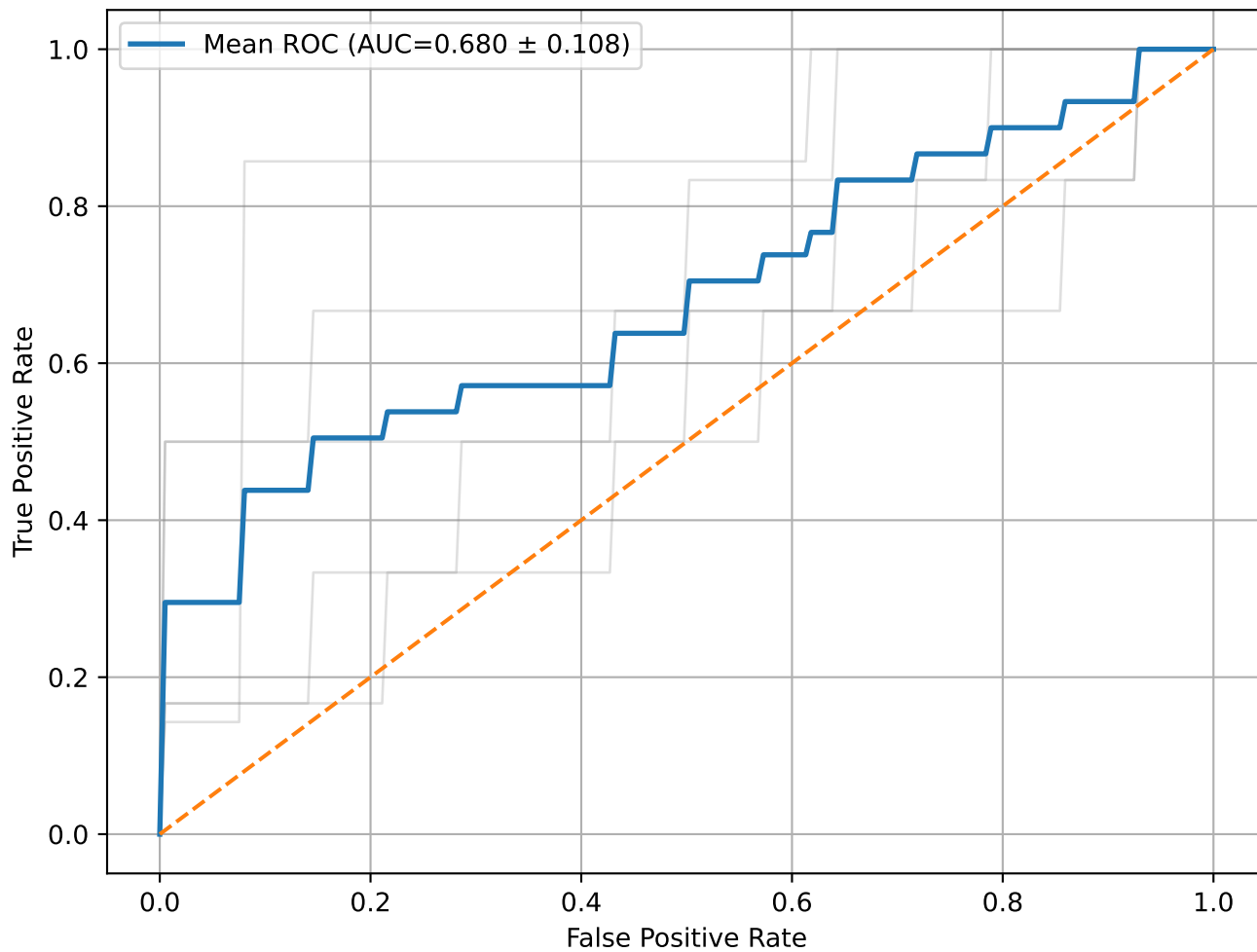
	precision	recall	f1-score	support	
0	0.76	1.00	0.87	13	
1	1.00	0.43	0.60	7	
accuracy				0.80	20
macro avg	0.88	0.71	0.73	20	
weighted avg	0.85	0.80	0.77	20	

Classification Report

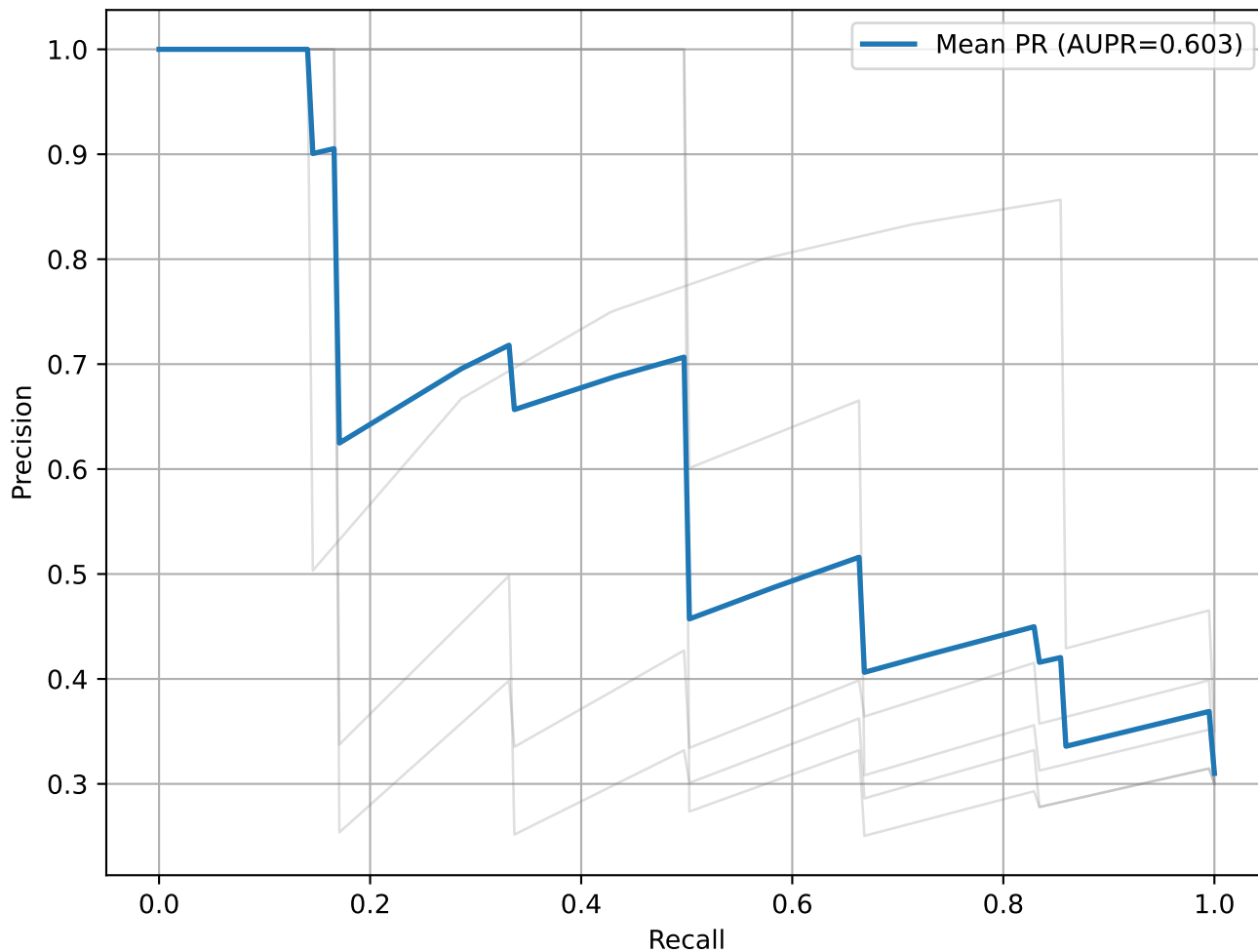
RF (last fold)

Classification Report - RF (last fold)

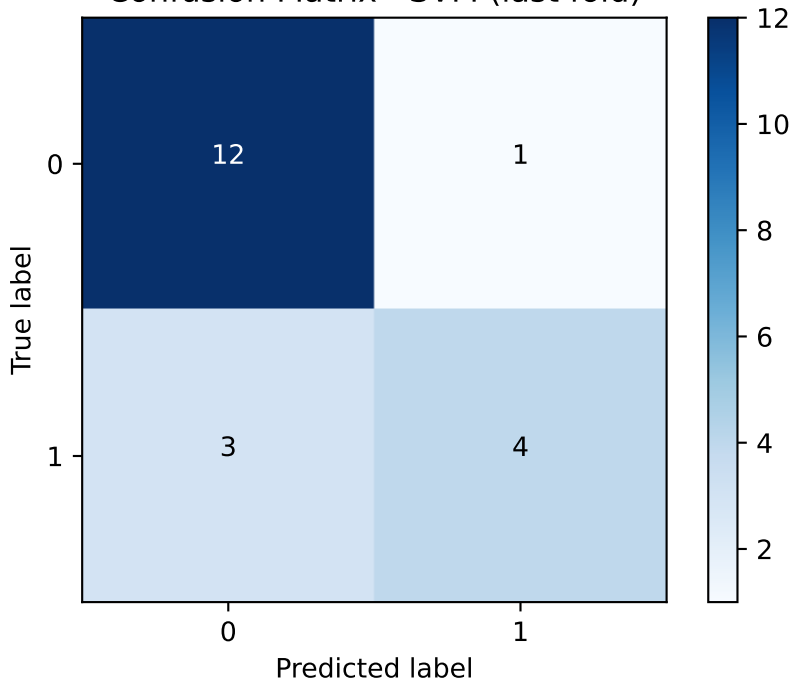
ROC - SVM



Precision-Recall - SVM



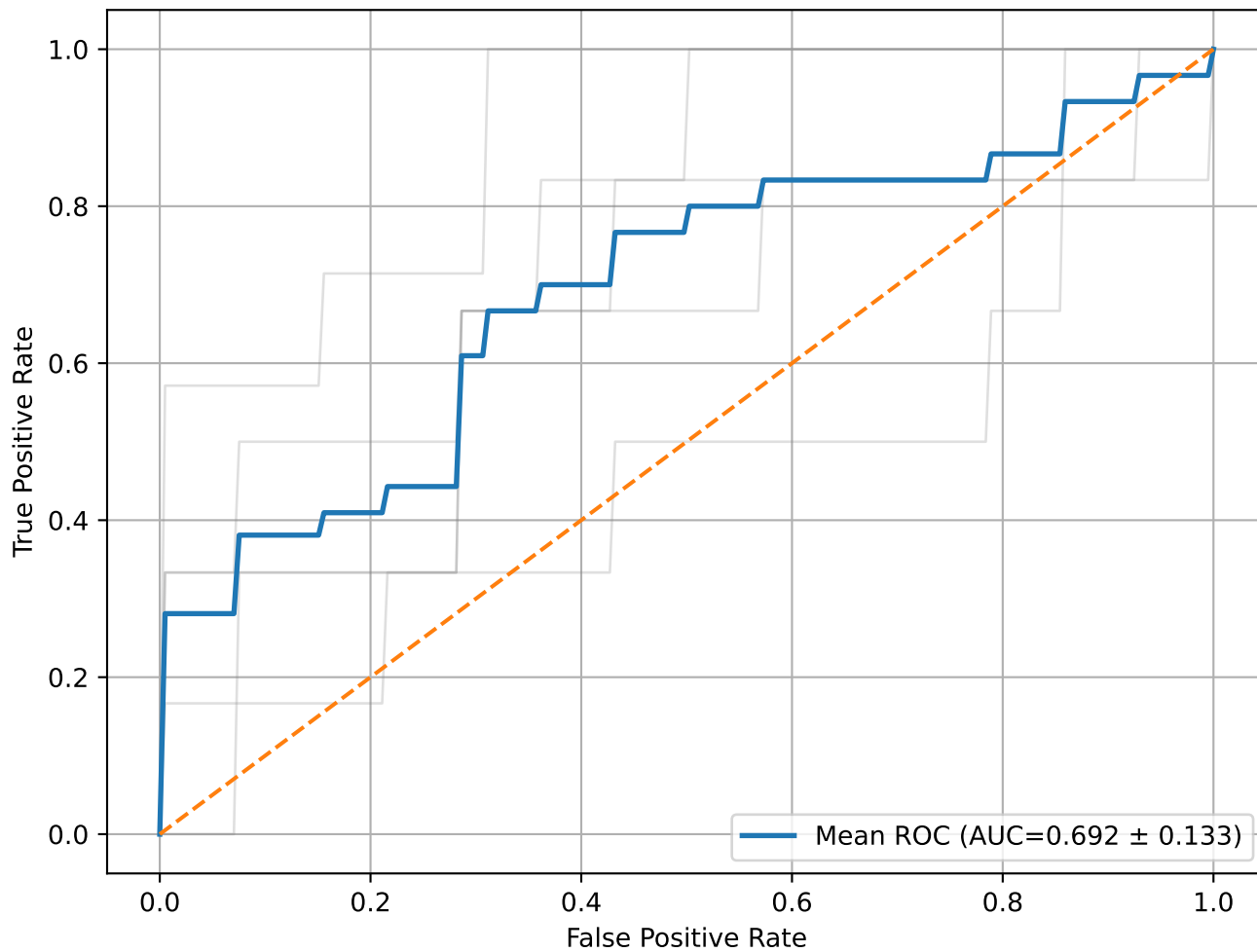
Confusion Matrix - SVM (last fold)



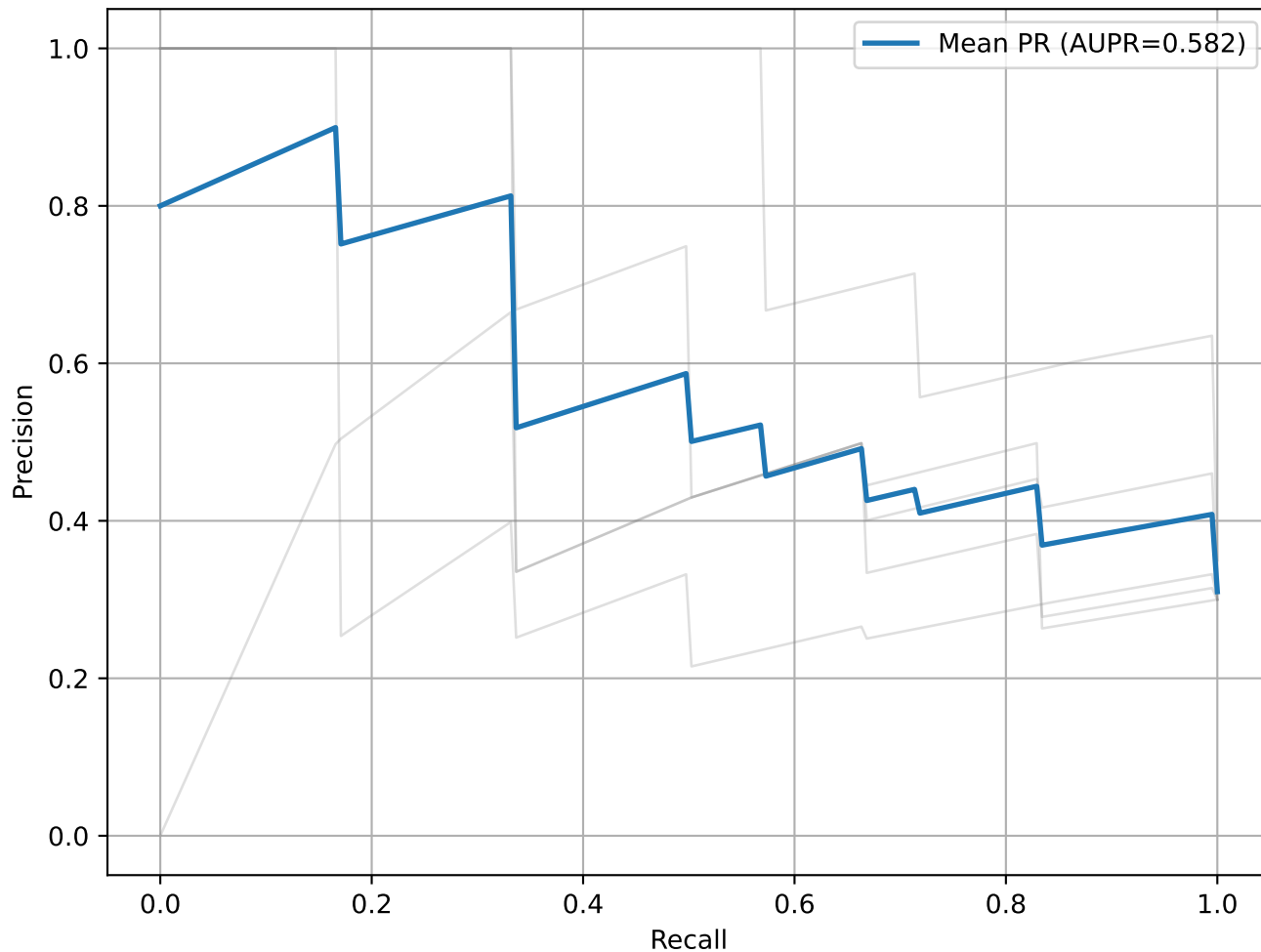
	precision	recall	f1-score	support
0	0.80	0.92	0.86	13
1	0.80	0.57	0.67	7
accuracy			0.80	20
macro avg	0.80	0.75	0.75	20
weighted avg	0.80	0.80	0.79	20

Classification Report - SVM (last fold)

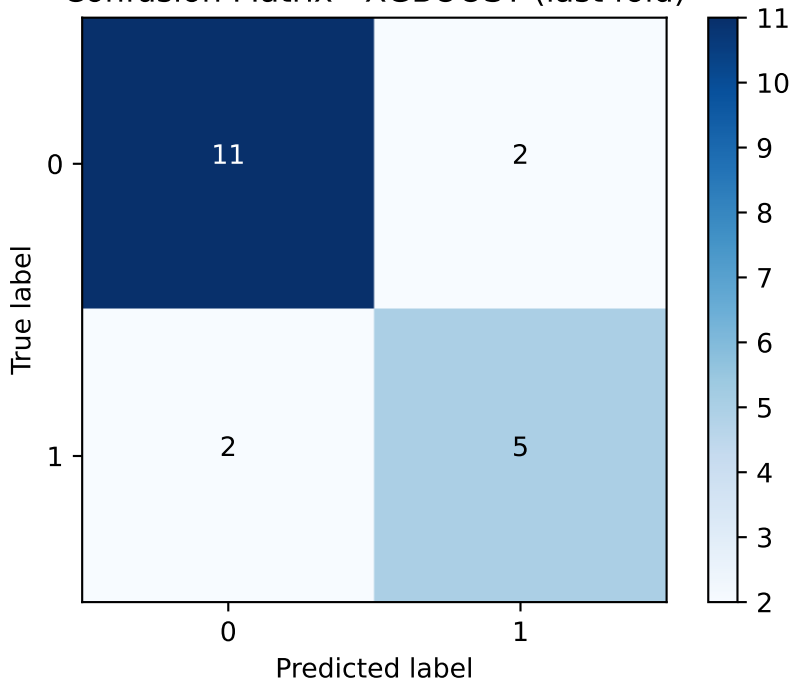
ROC - XGBOOST



Precision-Recall - XGBOOST



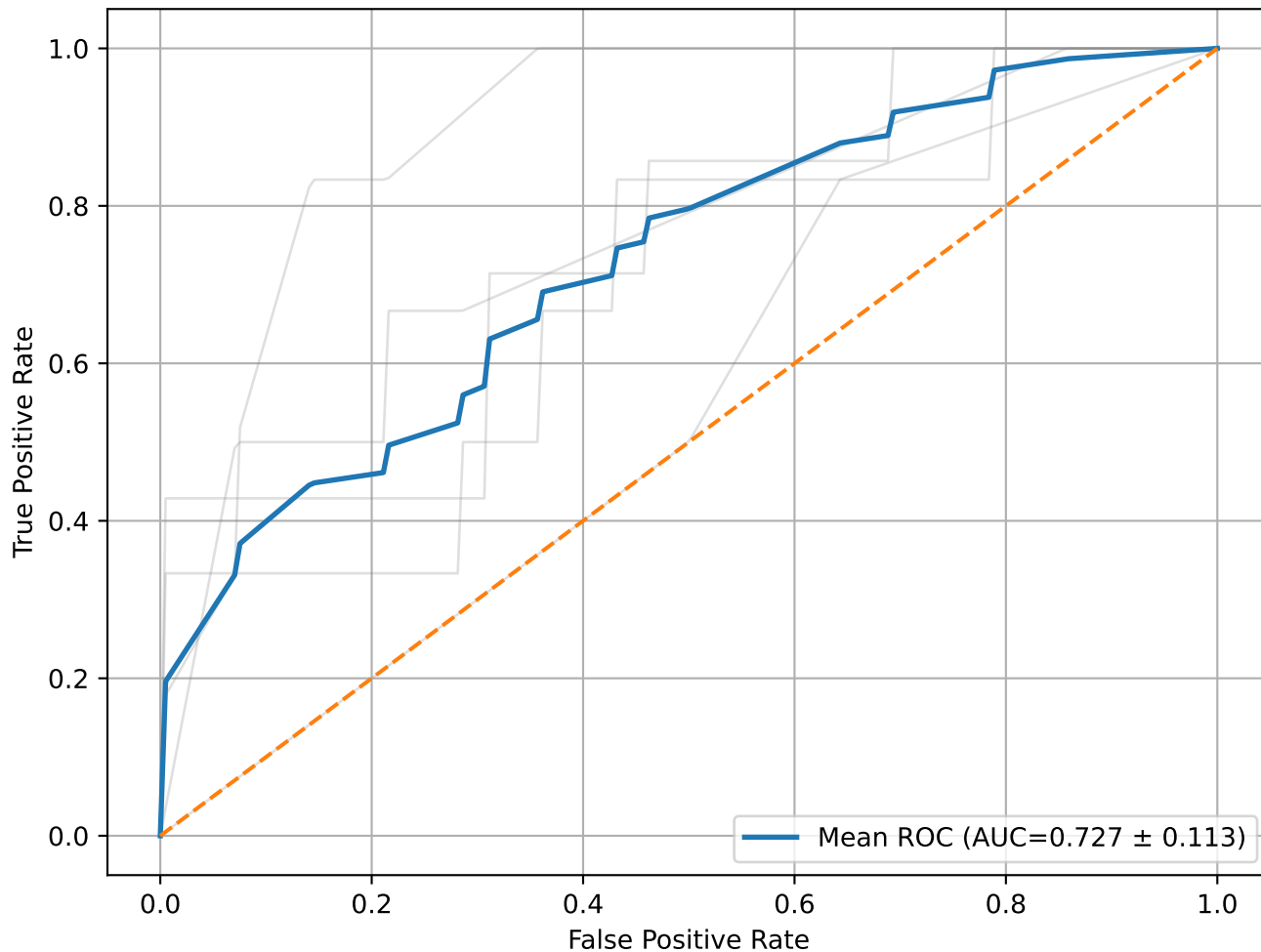
Confusion Matrix - XGBOOST (last fold)



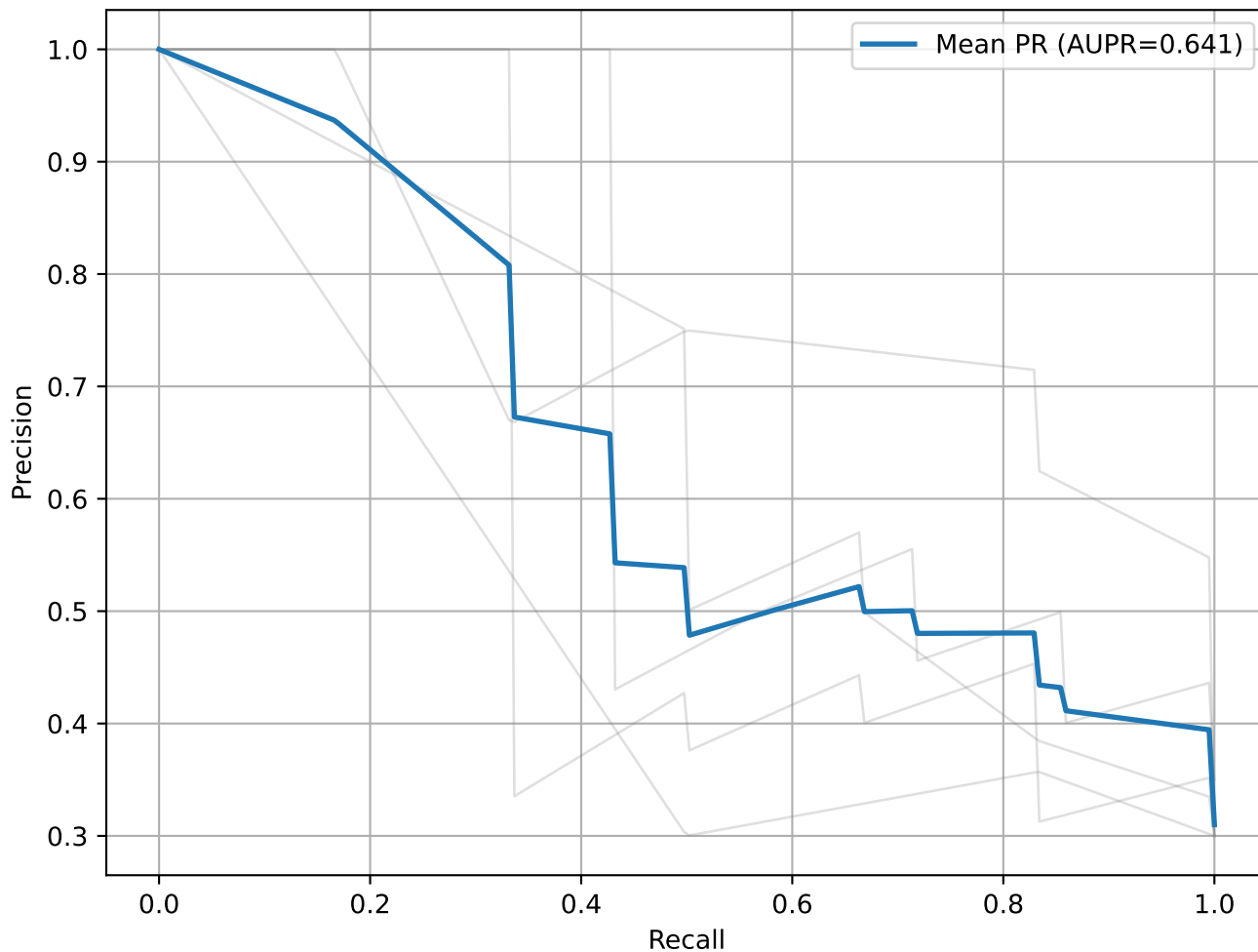
	precision	recall	f1-score	support
0	0.85	0.85	0.85	13
1	0.71	0.71	0.71	7
accuracy				0.80
macro avg	0.78	0.78	0.78	20
weighted avg	0.80	0.80	0.80	20

Classification Report - XGBOOST (last fold)

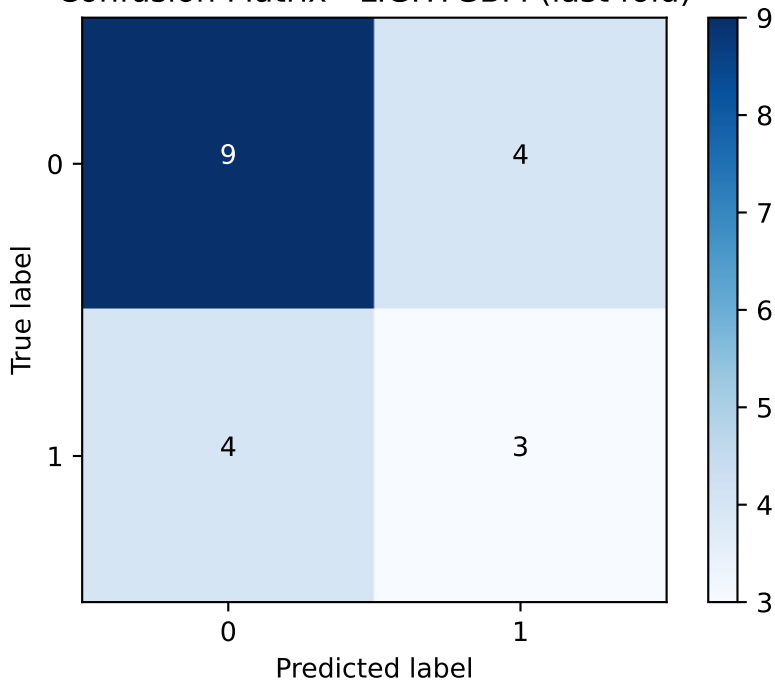
ROC - LIGHTGBM



Precision-Recall - LIGHTGBM



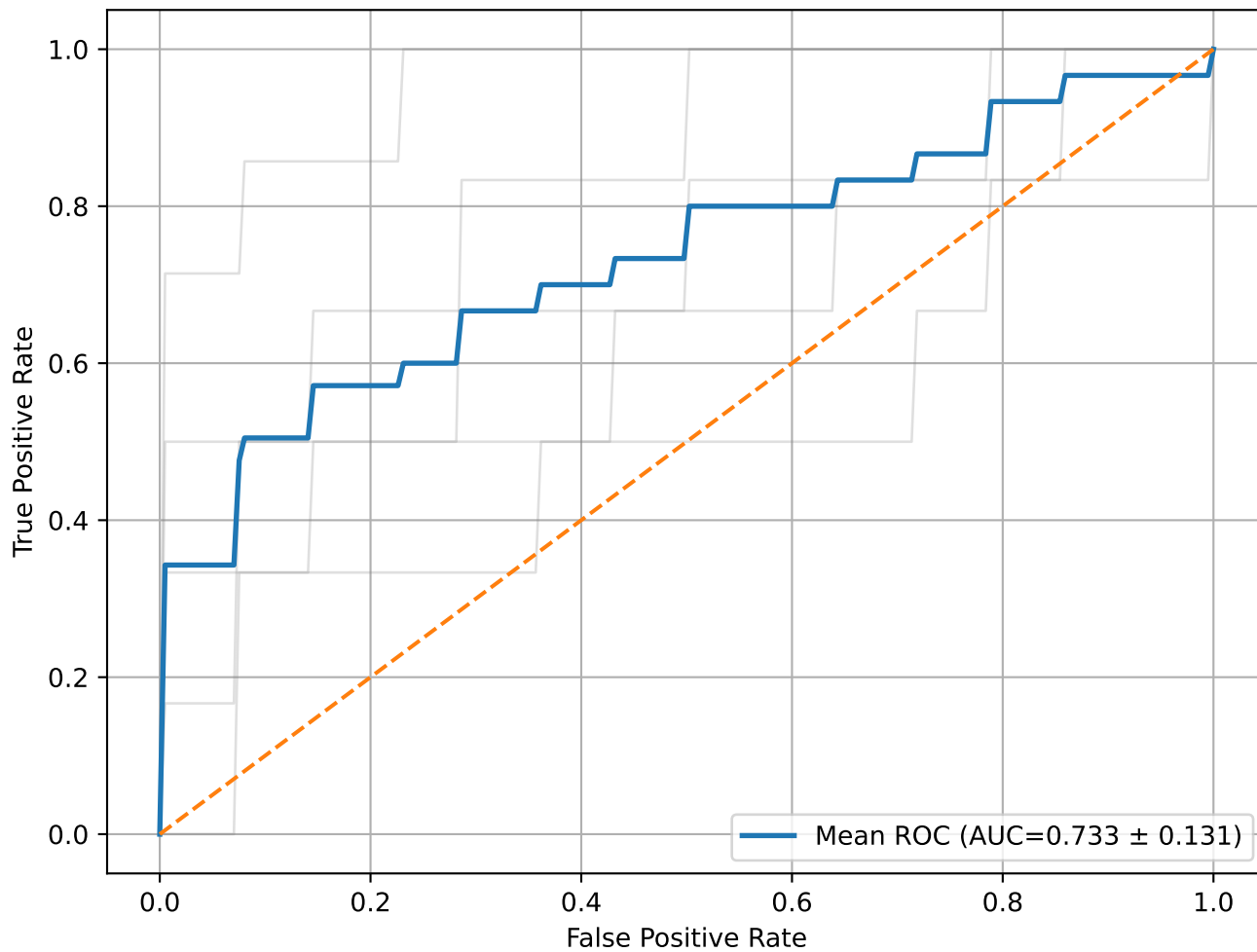
Confusion Matrix - LIGHTGBM (last fold)



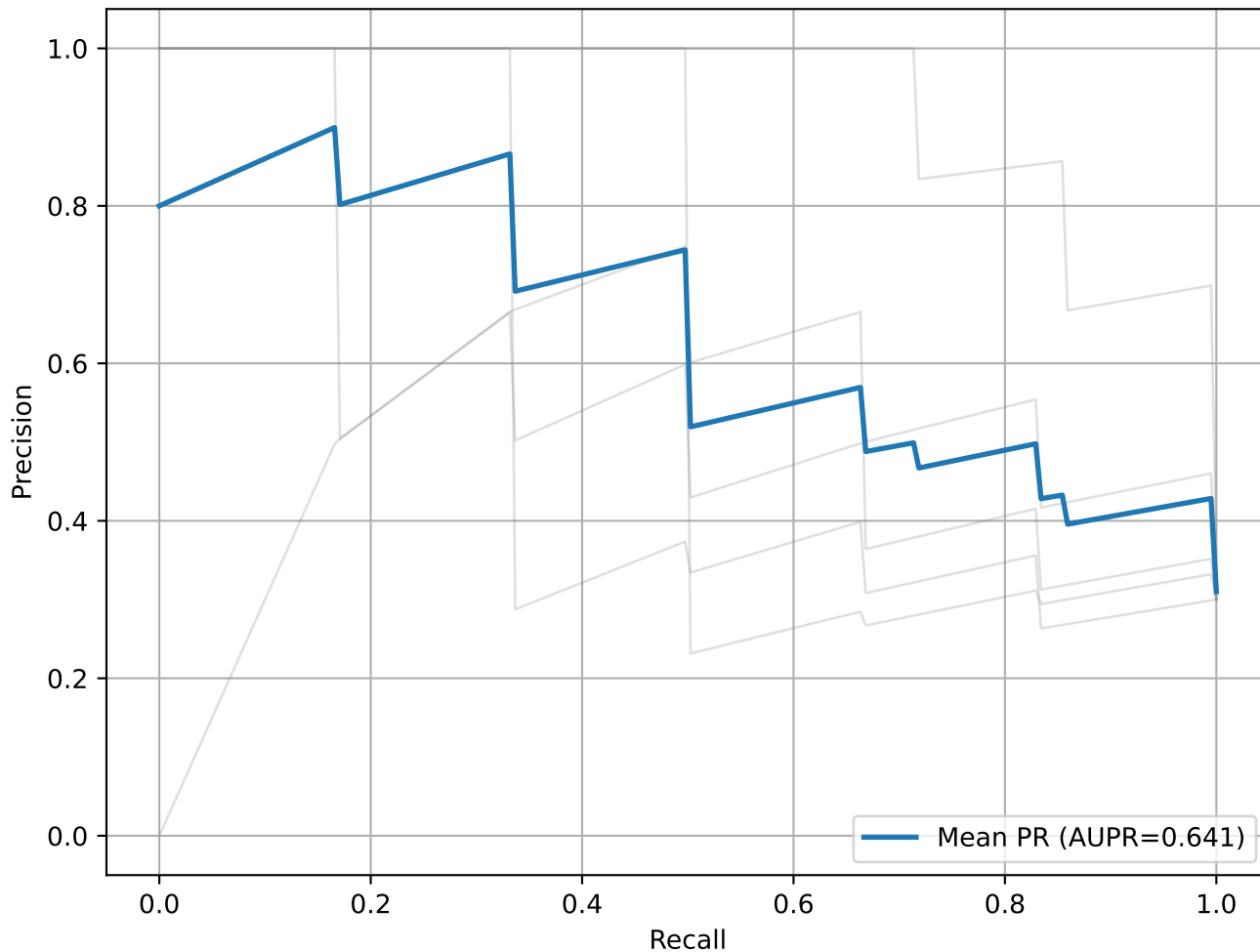
	precision	recall	f1-score	support
0	0.69	0.69	0.69	13
1	0.43	0.43	0.43	7
accuracy			0.60	20
weighted avg	0.60	0.56	0.58	20
weighted avg	0.60	0.60	0.60	20

Classification Report - LIGHTGBM (last fold)

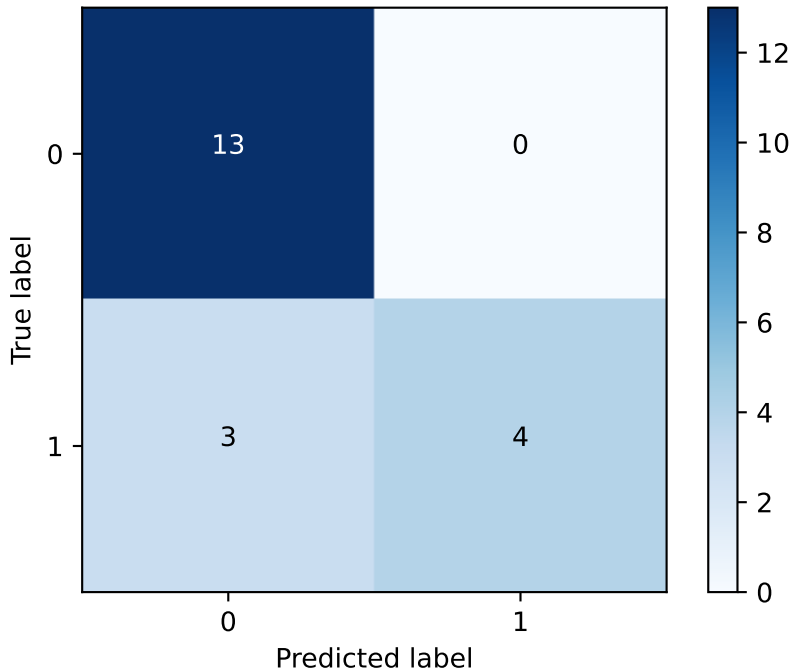
ROC - DNN



Precision-Recall - DNN



Confusion Matrix - DNN (last fold)



	precision	recall	f1-score	support
0	0.81	1.00	0.90	13
1	1.00	0.57	0.73	7
accuracy			0.85	20
macro avg	0.91	0.79	0.81	20
weighted avg	0.88	0.85	0.84	20

Classification Report - DNN (last fold)

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
dnn	0.734066	0.131341	0.641029	0.73	0.550490
logistic	0.728022	0.108910	0.635542	0.74	0.522424
lightgbm	0.728022	0.113378	0.640978	0.72	0.351746
xgboost	0.692308	0.133322	0.582027	0.70	0.449524
rf	0.691392	0.147834	0.617111	0.76	0.472092
svm	0.680952	0.107907	0.603163	0.74	0.363333